FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2019

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01 – 06, 19 – 24, 37 – 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 – 46) *contains 12 questions. Each question has* ***TWO (02)*** *matching lists:* ***LIST-I*** *and* ***LIST-II****.* ***FOUR*** *options are given representing matching of elements from* ***LIST-I*** *and* ***LIST-II****.* ***ONLY ONE*** *of these four options corresponds to a correct matching.*
   For each question, choose the option corresponding to the correct matching.
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

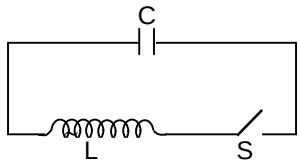
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. In the given circuit, initially the charge on capacitor is Q_0. At t = 0 switch S is closed. Which of the following statement(s) is/are correct?



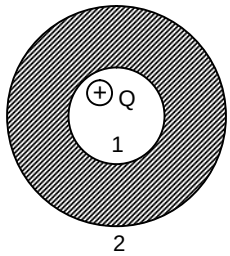
(A) Maximum current through the inductor is $\frac{Q_0}{\sqrt{LC}}$

(B) Charge on the capacitor is zero at $t = \frac{\pi}{2}\sqrt{LC}$

(C) Current through the inductor is maximum at $t = \frac{\pi}{2}\sqrt{LC}$

(D) Maximum magnetic energy can be stored in the inductor is $\frac{Q_0^2}{2C}$

2. The positive charge Q is placed in the cavity of the spherical conductor.

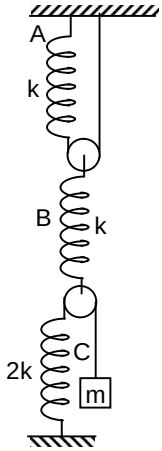


(A) the induced charge at the surface 1 is $-Q$ and on the surface 2 is +Q
(B) the surface charge density at inner surface is uniform
(C) the surface charge density at the outer surface is uniform
(D) the charge Q does not experience electric force if it is placed at the centre of the spherical cavity.

3. In case of earth:
(A) Potential is minimum at the centre of earth
(B) Potential is same, both at centre and infinity but not zero
(C) Potential is zero, both at centre and infinity
(D) Field is zero, both at centre and infinity

Space for Rough work

4. In the given figure, the block is attached with a system of three ideal springs A, B, C. The block is displaced by a small distance x from its equilibrium position vertically downwards and released. T represents the time period of small vertical oscillations of the block. Then (pulleys are ideal)



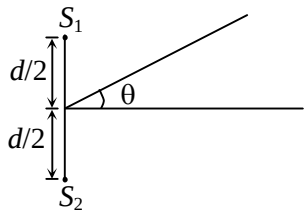
(A) $T = 2\pi\sqrt{\frac{11m}{2k}}$

(B) the deformation of the spring A is (2/11) times the displacement of the block.

(C) the deformation of the spring C is (1/11) times the displacement of the block.

(D) the deformation of the spring B is (4/11) times the displacement of the block.

5. In an interference arrangement, similar to Young's double slit experiment, the slits S_1 and S_2 are illuminated with coherent microwave sources, each of frequency 10^6 Hz. The sources are synchronized to have zero phase difference. The slits are separated by distance d = 150.0 m. The intensity $I_{(\theta)}$ is measured as a function of θ, where θ is defined as shown in the figure. If I_0 is maximum intensity, then $I_{(\theta)}$ for $0 \le \theta \le 90$ is given by



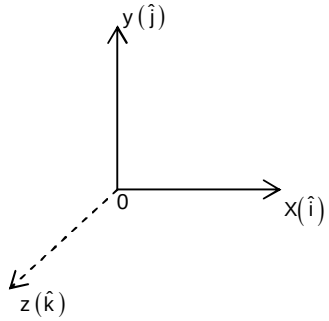
(A) $I_{(\theta)} = I_0$ for $\theta = 0°$

(B) $I_{(\theta)} = (I_0 / 2)$ for $\theta = 30°$

(C) $I_{(\theta)} = (I_0 / 4)$ for $\theta = 90°$

(D) $I_{(\theta)}$ is constant for all values of θ

6. A particle is projected from origin with velocity $\vec{u} = \left(\hat{i} + \hat{j} + \sqrt{2}\hat{k}\right)$ m/s. Horizontal surface lies in X – Y plane, then (take g = 10 m/sec^2)



(A) Time of flight = $\frac{\sqrt{2}}{5}$ sec

(B) horizontal range = $\frac{2}{5}$ m

(C) Maximum height $\frac{1}{10}$ m

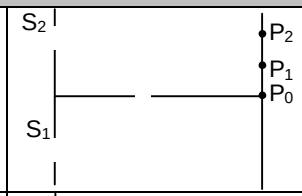
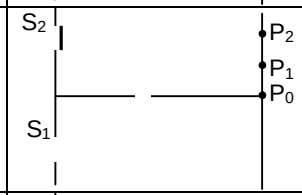
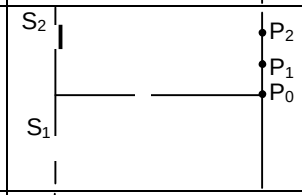
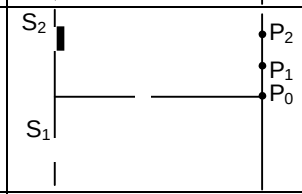
(D) Maximum height = $\frac{1}{5}$ m

Space for Rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

7. List-I shows four situations of standard young's double slit arrangement with screen placed far away from the slits S_1 and S_2. In each of these cases $S_1P_0 = S_2P_0$, $S_1P_1 - S_2P_1 = \lambda/4$ and $S_1P_2 - S_2P_2 = \lambda/3$, where λ is the wavelength of the light used. In the cases B, C and D, a transparent sheet of refractive index μ and thickness t is pasted on slit S_2. The thickness of the sheets are different in different cases. The phase difference between the light waves reaching a point P on the screen from the two slits is denoted by $\delta(p)$ and the intensity by I(p). Match each situation given in List-I with the statement(s) in List-II

List - I			List - II	
(P)			(1)	$\delta(P_0) = 0$
(Q)	$(\mu - 1)t = \lambda / 4$		(2)	$\delta(P_1) = 0$
(R)	$(\mu - 1)t = \lambda / 2$		(3)	$I(P_1) = 0$
(S)	$(\mu - 1)t = 3\lambda / 4$		(4)	$I(P_0) > I(P_1)$
			(5)	$I(P_2) > I(P_1)$

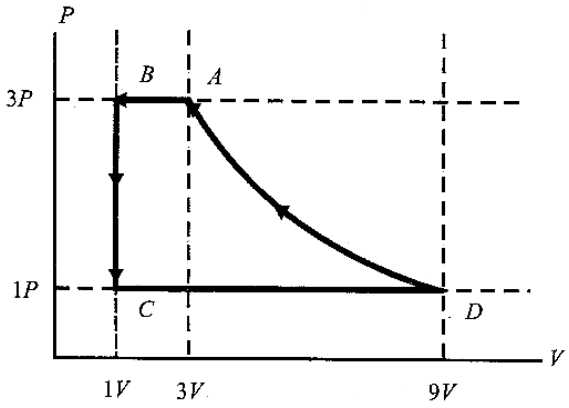
The correct option is:

(A) P→1,4 ; Q→3 ; R→5 ; S→3,5 (B) P→1,3 ; Q→4 ; R→5 ; S→3,4,5

(C) P→1,4 ; Q→2 ; R→3 ; S→3,4,5 (D) P→1,2 ; Q→5 ; R→4 ; S→1,5

Space for Rough work

8. One mole of a monatomic ideal gas is taken through a cycle ABCDA as shown in the P-V diagram. Column II gives the characteristics involved in the cycle. Match them with each of the processes given in Column I.



List – I		List – II	
(P)	Process A → B	(1)	Internal energy decreases
(Q)	Process B → C	(2)	Internal energy increases
(R)	Process C → D	(3)	Heat is lost
(S)	Process D → A	(4)	Heat is gained
		(5)	Work is done on the gas

The correct option is:

(A) P→1,3,5 ; Q→1,3 ; R→2,4 ; S→3,5 (B) P→1,2,5 ; Q→1,2 ; R→4 ; S→3,5

(C) P→1,5 ; Q→1,3 ; R→2,4 ; S→4,5 (D) P→3,5 ; Q→1,4 ; R→2,3 ; S→2,4

9. Average speed of a bus measured by six different students is given as follows:

Student number	1st	2nd	3rd	4th	5th	6th
Speed (m/s)	12.5 m/s	12.3 m/s	11.8 m/s	12.4 m/s	12.2 m/s	12.6 m/s

List -I		List -II	
(P)	Mean absolute error	(1)	± 0.2
(Q)	Mean value	(2)	12.3 m/s
(R)	Relative error	(3)	± 0.0163
(S)	Percentage error	(4)	± 1.63 %

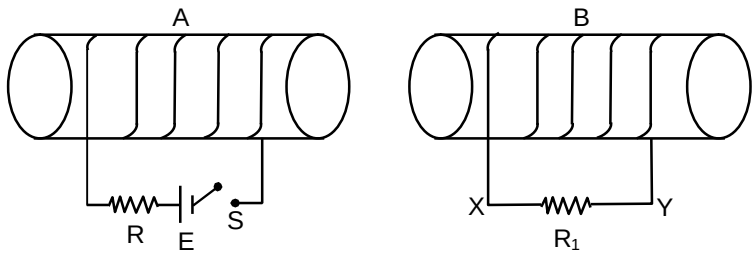
The correct option is:

(A) P→3 ; Q→1 ; R→2 ; S→4 (B) P→1 ; Q→2 ; R→3 ; S→4

(C) P→1 ; Q→3 ; R→4 ; S→2 (D) P→1 ; Q→4 ; R→3 ; S→2

Space for Rough work

10. In the circuit shown in figure two coils are arranged as shown. In List-I, some operation which is carried out in circuit I is mentioned and in List-II



about its effect. Match the entries of List – I with the entries of List–II.

List – I		List – II	
(P)	Switch S is closed (exclude small switching time)	(1)	Current in R_1 is form X to Y.
(Q)	Switch S is closed for long time then it opened for this transition time.	(2)	Current in R_1 is form Y to X
(R)	If coil A is move, perpendicular to B, switch S is closed.	(3)	No current is flowing through R_1.
(S)	The battery of constant emf is replaced by a varying emf and switch S is closed.	(4)	Current in R_1 can be from X to Y or form Y to X.
		(5)	Coil B will attract coil A

The correct option is:

(A) P→2 ; Q→1, 3 ; R→5 ; S→4 (B) P→3 ; Q→2 ; R→2, 5 ; S→3,4

(C) P→3 ; Q→4 ; R→1, 2 ; S→4 (D) P→1 ; Q→2, 5 ; R→2, 5 ; S→4

SECTION – D

(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. Two blocks of masses 2kg and 5kg are connected by a light string passing over a frictionless pulley. Find the tension (in Newton) in the cord connecting the masses. (G = 9.8 M/S^2)

12. A bullet losses 19% of its kinetic energy when passes through an obstacle. Find the fractional change in its speed

Space for Rough work

13. A 10 m wide spacecraft moves through the interstellar space at a speed 3×10^6 m/s. A magnetic field $B = 3 \times 10^{-9}$ T exists in the space in a direction perpendicular to the plane of motion. Treating the spacecraft as a conductor, calculate the emf induced across its width. (in volt)

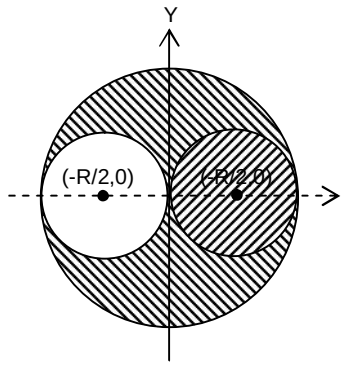
14. The refractive index for water w.r.t. air for a sound wave if the velocities of sound in water and air are 1440 ms^{-1} and 340ms^{-1}, is approximately,

15. Two glass plates are separated by water. If surface tension of water is 75 dynes per cm and area of each plate wetted by water is 8 cm^2 and the distance between the plates is 0.12 mm, then the force (in newton) applied to separate the two plates is

16. An organ pipe P_1 closed at one end vibrating in its first overtone and another pipe P_2 open at the both ends vibrating in its third overtone are in resonance with a given tuning fork. The ratio of the length of P_1 to that of P_2 is $\frac{3}{n}$ then n = ?

17. Gravitational filed in a region is given by $\vec{E} = \left(3\hat{i} - 4\hat{j}\right)$ N/kg. Find out the work done (in J) in displacing a particle of mass 1 unit by 1m along the line 4y = 3x + 9?

18. Figure shows a uniform disc of radius R from which a hole of radius R/2 has been cut out from left of the centre and is placed on right of the centre of disc. The centre of mass of resulting disc is at a distance R/n from centre of uniform disc. Find the value of n



Space for Rough work

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. The paramagnetic species out of the following is/are

(A) O_2^+ (B) N_2^+

(C) O_2^{2-} (D) N_2^{2-}

20. The following data are given for the reaction

$X(g) + Y(g) \longrightarrow$ Product

No. of expt.	Initial concentration of 'X' in 'M'	Initial concentration of 'Y' in 'M'	Rate of reaction in M s^{-1} unit
1.	0.1	0.1	2×10^{-8}
2.	0.2	0.2	4×10^{-8}
3.	0.4	0.2	8×10^{-8}

Choose the correct statement(s) from the following

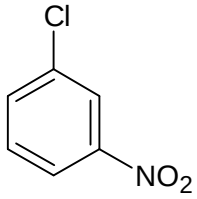
(A) The order of reaction w.r.t X is one

(B) The overall order of reaction is one

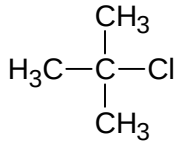
(C) The rate constant of the reaction is 2×10^{-7} s^{-1}

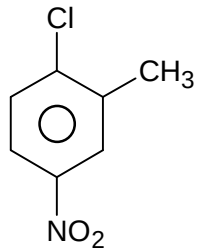
(D) The half-life of 'Y' does not depend on its concentration

21. In which of the following compound(s) the C – Cl bond is shorter than that in C_2H_5Cl?

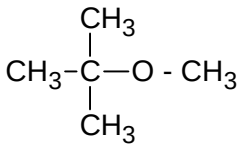
(A) 

(B) $CH_2 = CH - Cl$

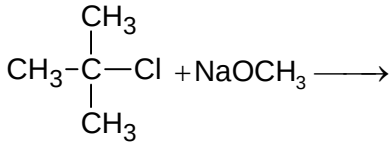
(C) 

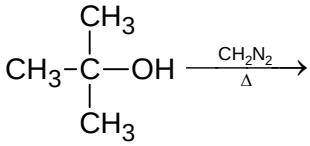
(D) 

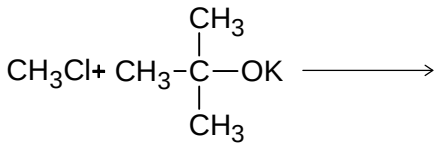
Space for Rough work

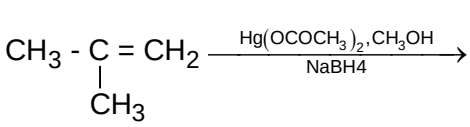
22. 

Which of the following reaction(s) form the above compound?

(A) 

(B) 

(C) 

(D) 

23. Which of the following statements is/are true about an azeotropic mixture?
(A) An azeotropic mixture boils at constant temperature
(B) The composition of an azeotropic mixture changes on distillation
(C) An azeotropic solution of two liquids has a boiling point lower than that of either of them when it shows positive deviation from the Raoult's law
(D) An azeotropic solution of two liquids has a boiling point higher than that of either of them if it shows positive deviation from Raoult's law

24. Which of the following statements is / are correct
(A) The coordination number of each type of ion in CsCl crystal is 8
(B) A metal that crystallizes in bcc structure has a coordination number of 12
(C) A unit cell of an ionic crystal shares some of its ions with other unit cells
(D) Length of the unit cell in NaCl is 552 pm ($r_{Na^+} = 95pm; r_{Cl^-} = 181pm$)

Space for Rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

25. Match the bases mentioned in List - I with their properties mentioned in List - II.

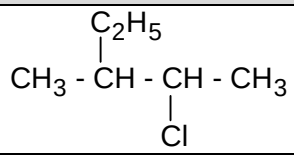
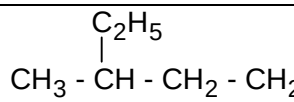
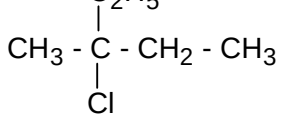
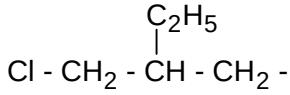
List – I		List – II	
(P)	LiOH	(1)	Undergoes dehydration on heating to produce normal oxide
(Q)	NaOH	(2)	Used as an antacid
(R)	KOH	(3)	Used to prepare soft soaps
(S)	$Mg(OH)_2$	(4)	Absorbs CO_2 to form a compound which hydrated form is called washing soda

The correct option is

(A) P → 3; Q → 4; R → 1; S → 2 (B) P → 1; Q → 3,4; R → 3; S → 1, 2

(C) P → 2,3; Q → 1; R → 3; S → 4 (D) P → 1; Q → 3,4; R → 2; S → 2,4

26. Match the alkyl chlorides mentioned in List-I with the nature of their elimination reaction products mentioned in List- II.

List – I		List – II	
(P)		(1)	One product shows geometrical isomerism if formed through E_2 mechanism
(Q)		(2)	One product shows geometrical isomerism if formed through E_1 mechanism
(R)		(3)	One product shows optical isomerism if formed through E_2 mechanism
(S)		(4)	One product shows optical isomerism if produced through E_1 mechanism

The correct option is

(A) P → 3; Q → 4; R → 1; S → 2, 3, 4

(B) P → 2,3; Q → 1,4; R → 3; S → 1, 2

(C) P → 1,2,3,4; Q → 2,3,4; R → 1,2; S → 2

(D) P → 1; Q → 3,4; R → 2; S → 2,4

Space for Rough work

27. Match the compounds mentioned in List -I with their characteristics mentioned in List-II

List – I		List – II	
(P)	CH_3COCl	(1)	Forms CH_3COOH as only or one of the products on hydrolysis
(Q)	$CH_3COOC_2H_5$	(2)	Forms two organic compounds on hydrolysis
(R)	$(CH_3CO)_2O$	(3)	Hydrolysis is irreversible in both acidic and basic medium
(S)	CH_3CONH_2	(4)	Undergoes the easiest nucleophilic substitution reaction among the four compounds

The correct option is

(A) P → 3; Q → 4; R → 1; S → 2 (B) P → 2,3; Q → 3,4; R → 3; S → 1, 2

(C) P → 1,3,; Q → 1,,4; R → 1,2; S → 2 (D) P → 1, 4; Q → 1,2; R → 1; S → 1,3

28. Match the precipitates of the compounds listed in List -I with the solvent(s) listed in List-II.

List – I		List – II	
(P)	$Zn(OH)_2$ precipitate dissolves in	(1)	Potassium cyanide
(Q)	$Cr(OH)_3$ precipitate dissolves in	(2)	Ammonia solution
(R)	AgCl precipitate dissolves in	(3)	Sodium hydroxide
(S)	CuS precipitate dissolves in	(4)	Hydrochloric acid

The correct option is

(A) P → 1, 2, 3, 4; Q → 2, 3, 4; R → 1, 2; S → 1

(B) P → 2,3; Q → 1, 2; R → 3; S → 1, 2

(C) P → 1,3,; Q → 1,,4; R → 1,2; S → 2

(D) P → 2, 3, 4; Q → 3; R → 1; S → 4

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. The following two reactions attain equilibrium in a container by taking in initially only $FeSO_4$, and O_2 gas is passed then

$2FeSO_4(s) \rightleftharpoons Fe_2O_3(s) + SO_2(g) + SO_3(g); K_{P_1} = 900\,cm^2$ of Hg

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g); K_{P_2} = 25 \times 10^{-4}\,cm^{-1}$ of Hg

What is the equilibrium partial pressure of $O_2(g)$ in cm of Hg unit?

30. The molar mass of the salt MX_2 is 208.3 g. The solubility product of MX_2 is 4×10^{-3}. How many gram of the salt should be dissolved in water to form 10 litre saturated solution?

31. $KO_2(s) + CO_2(g) \longrightarrow P(s) + Q(g)$

$\xrightarrow{H_2O} R(soln) + S(soln) + Q(g)$

$\xrightarrow[\text{Starch}]{KI\,(soln)/H^+}$ Blue solution

What is the molar mass in g of the heaviest unknown product (out of P, Q, R and S) in above reactions?

32. $(P) \xrightarrow[\text{Alcohol, }\Delta]{KOH} (Q) \xrightarrow{Br_2/CCl_4} (R)$

(Alkyl chloride)

$CH_3 - \underset{\diagdown\,\diagup\ O}{CH} - CH_2 \xleftarrow[-Br^-]{KOH(1\,eq)} (S) \xleftarrow[H_2O,\ S_N1]{KOH(1\,eq)}$ (R)

How many grams of (P) is required to form one mole of 2, 3-dimethylbutane upon treatment with sodium metal in dry ether?

Space for Rough work

33. $HO-CH_2-\underset{\underset{OH}{|}}{CH}-CH_2COOH \xrightarrow{SOCl_2(excess)} Product$

What will be the molar mass in gram of the organic product of above reaction?

34. Cr(III) forms an octahedral complex with NH_3 and Cl ligands. The van't Hoff factor of the complex is 2 when it is 100% ionized in water. The complex ion shows geometrical isomerism. What is the molar mass of the complex in gram unit?

35. Calculate the reversible work done at 5000 K in joule unit by compressing one mole of an ideal gas to one-tenth $\left(\frac{1}{10}\right)$ of its original volume in an isothermal process?

36. The molar conductance at infinite dilution of $CaCl_2$, $Ca(CH_3COO)_2$ and HCl are respectively 271.6, 200.8 and 425.95 S cm^2 mol^{-1}. What will be the molar conductance of CH_3COOH in S cm^2 mol^{-1} unit?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. The value of the expression $\tan\frac{\pi}{7}+2\tan\frac{2\pi}{7}+4\tan\frac{4\pi}{7}+8\cot\frac{8\pi}{7}$ is equal to:

(A) $\operatorname{cosec}\frac{2\pi}{7}+\cot\frac{2\pi}{7}$ (B) $\tan\frac{\pi}{14}-\cot\frac{\pi}{14}$

(C) $\dfrac{\sin\frac{2\pi}{7}}{1-\cos\frac{2\pi}{7}}$ (D) $\dfrac{1+\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}}{\sin\frac{\pi}{7}+\sin\frac{2\pi}{7}}$

38. Which of the following pair(s) of function have same graphs?

(A) $f(x)=\dfrac{\sec x}{\cos x}-\dfrac{\tan x}{\cot x}, g(x)=\dfrac{\cos x}{\sec x}+\dfrac{\sin x}{\operatorname{cosec} x}$

(B) $f(x)=\operatorname{sgn}\left(x^2-4x+5\right), g(x)=\operatorname{sgn}\left(\cos^2 x+\sin^2\left(x+\dfrac{\pi}{3}\right)\right)$ where sgn denotes signum function.

(C) $f(x)=e^{\ln\left(x^2+3x+3\right)}, g(x)=x^2+3x+3$

(D) $f(x)=\dfrac{\sin x}{\sec x}+\dfrac{\cos x}{\operatorname{cosec} x}, g(x)=\dfrac{2\cos^2 x}{\cot x}$

39. Let $f(x)=\cos^{-1}\left(2x^2-1\right)$ then:

(A) $f(x)$ is continuous in $[-1, 1]$

(B) $f(x)$ is derivable in (–1, 1)

(C) range of $f(x)$ is $[0, \pi]$

(D) derivative of $f(x)$ w.r.t. $\sin^{-1}x$ at $x=\dfrac{1}{2}$ is 2

Space for Rough work

40. Suppose $I_1 = \int_0^{\pi/2} \cos\left(\pi \sin^2 x\right) dx; I_2 = \int_0^{\pi/2} \cos\left(2\pi \sin^2 x\right) dx$ and $I_3 = \int_0^{\pi/2} \cos\left(\pi \sin x\right) dx$, then

(A) $I_1 = 0$ (B) $I_2 + I_3 = 0$

(C) $I_1 + I_2 + I_3 = 0$ (D) $I_2 = I_3$

41. Consider the function $f(x) = \left[\cos\left(\tan^{-1}\left(\sin\left(\cot^{-1} x\right)\right)\right)\right]^2$. Which of the following is correct?

(A) Range of f is (0, 1)

(B) f is even

(C) $f'(0) = 0$

(D) The line y = 1 is asymptotes to the graph $y = f(x)$

42. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx}(\sin x) - y\cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \to 0$ as $x \to \infty$ then the statement which is correct is:

(A) $\lim_{x \to 0} f(x) = 1$ (B) $\int_0^{\pi/2} f(x)dx$ is less than $\frac{\pi}{2}$

(C) $\int_0^{\pi/2} f(x)dx$ is greater than unity (D) $f(x)$ is an odd function

Space for Rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

43. Match the following

List-I		List-II	
(P)	If m and n are positive integers satisfying $1+\cos 2\theta+\cos 4\theta+\cos 6\theta+\cos 8\theta+\cos 10\theta = \frac{\cos m\theta.\sin n\theta}{\sin\theta}$, then $(m+n)$ is equal to	(1)	9
(Q)	The minimum value of the expression $\frac{9x^2+\sin^2 x+4}{x\sin x}$ for $x\in(0,\pi)$ is	(2)	10
(R)	Let $f(x)=11-8\sin x-2\cos^2 x$. If the maximum and minimum value of $f(x)$ are denoted by M and m respectively, then $\frac{M+8}{m}$ has the value equal to	(3)	11
(S)	If $\tan 9\theta=\frac{3}{4}$ (where $0<\theta<\frac{\pi}{18}$), then the value of $(3\operatorname{cosec} 3\theta-4\sec 3\theta)$ is equal to	(4)	12
		(5)	13
		(6)	10

The correct option is

(A) P → 3; Q → 4; R → 1; S → 2 (B) P → 1; Q →4; R →3; S → 4

(C) P → 1; Q → 1; R → 3; S → 6 (D) P → 1; Q → 1; R → 5; S → 4

Space for Rough work

44. Match the following

List-I		List-II	
(P)	Let f be a real valued differentiable function on f such that $f'(1)=6$ and $f'(2)=2$. Then $\lim_{h\to 0}\frac{f(3\cos h+4\sin h-2)-f(1)}{f(3e^h-5\sec h+4)-f(2)}$ is equal to	(1)	4
(Q)	For $a>0$, let $f:[-4a, 4a]\to R$ be an even function such that $f(x)=f(4a-x)\forall x\in[2a, 4a]$ and $\lim_{h\to 0}\frac{f(2a+h)-f(2a)}{h}=4$ then $\lim_{h\to 0}\frac{f(h-2a)-f(-2a)}{2h}$ is equal to	(2)	5
(R)	Suppose f is a differentiable function on R. Let $F(x)=f(e^x)$ and $G(x)=e^{f(x)}$. If $f'(1)=e^3$ and $f(0)=f'(0)=3$, then $\frac{G'(0)}{F'(0)}$ is equal to	(3)	3
(S)	Let $f(x)=\text{Max.}(\cos x, x, 2x-1)$ where $x\geq 0$. Then number of points of non – differentiability	(4)	2
		(5)	1
		(6)	0

The correct option is

(A) P → 1; Q → 1; R → 3; S → 6 (B) P → 1; Q →4; R →3; S → 4

(C) P → 1; Q → 1; R → 3; S → 6 (D) P → 1; Q → 1; R → 6; S → 2

Space for Rough work

45. Consider $f(x) = |\ln|x|| - kx^2$, $x \neq 0$. Match the column – I with the value of k in column – II

List-I		List-II	
(P)	$f(x) = 0$ has two distinct solutions	(1)	$k = 0$
(Q)	$f(x) = 0$ has four distinct solutions	(2)	$k = \frac{1}{2e}$
(R)	$f(x) = 0$ has six distinct solutions	(3)	$k \in \left(\frac{1}{2e}, \infty\right)$
(S)	$f(x) =$ has no solution	(4)	$k \in (-\infty, 0)$
		(5)	$k \in \left(0, \frac{1}{2e}\right)$
		(6)	$k = \frac{1}{e}$

The correct option is

(A) P → 2; Q → 3; R → 5; S → 4 (B) P → 3; Q →5; R →4; S → 5

(C) P → 1; Q → 2; R → 5; S → 4 (D) P → 1; Q → 1; R → 5; S → 2

46. Match the following

List-I		List-II	
(P)	Let $f: R \to R$ defined as $f(x) = e^{\text{sgn}(x)} + e^{x^2}$, where sgn x denotes signum function of x, then $f(x)$ is	(1)	Odd
(Q)	Let $f: (-1, 1) \to R$ defined as $f(x) = x\left[x^4\right] + \frac{1}{\sqrt{1-x^2}}$ where [x] denotes greatest integer less than or equal to x, then $f(x)$ is	(2)	Even
(R)	Let $f: R \to R$ defined as $f(x) = \frac{x(x+1)\left(x^4+1\right) + 2x^4 + x^2 + 2}{x^2 + x + 1}$, then $f(x)$ is	(3)	Neither odd nor even
(S)	Let $f: R \to R$ defined as $f(x) = x + 3x^3 + 5x^5 + ... + 101x^{101}$ then $f(x)$ is	(4)	One – One
		(5)	Many – one
		(6)	Constant

The correct option is

(A) P → 2, 3; Q → 3,4; R → 5,1; S → 4, 2 (B) P → 3, 2; Q →5. 4; R →3, 4; S →5, 6

(C) P → 1, 2; Q → 2, 6; R → 5, 4; S → 2, 4 (D) P → 3, 5; Q → 2, 5; R → 3, 5; S → 1, 4

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. If solution of $(x\cos y - y\sin y)dy + (x\sin y + y\cos y)dx = 0$ is $\ell n(x\sin y + y\cos y - \sin y) = -4kx + c$ then k is

48. Given sequence of numbers $x_1, x_2, x_3, ...x_{2005}$ which satisfy, $\frac{x_1}{x_1+1} = \frac{x_2}{x_2+3} = \frac{x_3}{x_3+5} ..... \frac{x_{2005}}{x_{2005}+2009}$ where $x_1 + x_2 + ...x_{1005} = 2010$ then $\frac{1005}{164}x_{21}$ is

49. If roots of $x^3 + ax^2 + bx - \frac{1}{9} = 0$ are in A.P. the $\frac{121}{64}(2a^3 - 9ab + 1)$ is _____.

50. A bag contain N marbles (N < 200) of 4 colours: Red, White and Green. Following events are equally likely
(i) Selection of 4 Red marbles
(ii) Selection of 1 White and 3 Red marbles
(iii) Selection of 1 White, 1 Blue and 2 Red marbles
(iv) Selection of 1 marble of each color
If S is sum of digits of maximum possible value of N then $\frac{7S}{500}$ is

51. Let A and B be two non singular matrices of order 2 such that $9A^2B - 6AB + B = 9\left\{B - \begin{bmatrix} 2 & -3 \\ 2 & 1 \end{bmatrix}\right\}$. If $B = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}$ and Trace (A) $(A) = \frac{10}{3}$ then $\frac{|A|}{4}$ is

Space for Rough work

52. If $\sum_{k=0}^{100}\left(\frac{k}{k+1}\right)^{100}C_k = \frac{a\left(2^{100}\right)+b}{c}$ where $a,b,c \in N$, then find the least value of $\frac{(a+b+c)}{670}$.

53. If the sum of an infinite geometric series, whose first term is the limit of the function $f(x)=\left(\frac{9}{1-x^9}-\frac{7}{1-x^7}\right)$ as $x \to 1$ and the common ratio is the limit of function $g(x)=\frac{\tan x-\sin x}{\ln^3(1+x)}$ as $x \to 0$, is S then $\frac{7S}{100}$ is

54. If $f:R \to R$ be an injective mapping and p, q, r are non – zero distinct real quantities satisfying $f\left(\frac{p}{r}\right)=f\left(\frac{p-q}{q-r}\right)$ and $f\left(\frac{q}{r}\right)=f\left(\frac{r}{p}\right)$

If the graph of $g(x)=px^2+qx+r$ passes through M (1, 6) then find the value of $\frac{q}{32}$.

Space for Rough work